Title: LW-Diffusion: An Optimized Lightweight AIGC Text-to-Image Diffusion Model for Resource-Constrained Edge Devices

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Abstract

Generative-AI text-to-image diffusion models possess powerful multi-modal content-generation capabilities. Nevertheless, vanilla diffusion models contain massive parameter volumes and demand numerous iterative sampling steps; most of them can only be deployed on high-performance cloud GPU servers, which makes direct local execution infeasible for resource-constrained edge-side hardware such as mobile smartphones and embedded IoT devices. This paper proposes **LW-Diffusion**, an integrated lightweight-optimization solution for AIGC text-to-image diffusion models targeting memory- and compute-limited edge-side devices. The proposed scheme combines structured model pruning, mixed-precision quantization, and few-step-sampling knowledge distillation. Importance-aware structured pruning is performed on the U-Net backbone network to eliminate redundant convolution channels. Differentiated mixed-precision quantization assigns distinct quantization bit-widths for different network layers according to per-layer sensitivity characteristics. A few-step-sampling knowledge-distillation strategy transfers the generative capability from a high-step teacher diffusion model into a low-step student model, drastically reducing inference-sampling iteration overhead. Moreover, an edge-aware multi-component joint-loss function is designed to mitigate image-generation-quality degradation induced by pruning, quantization and distillation operations. Evaluated on the public COCO-2014 text-to-image dataset, LW-Diffusion compresses model parameters down to 31.7 % of original Stable Diffusion v1.5, reduces inference sampling steps from 50 down to 8. Average single-image inference time under regular edge-side CPU-only simulation environment decreases from 28.7 s to 4.2 s, achieving an FID score of 23.67. It outperforms comparative baselines including naive direct pruning and naive global quantization with only moderate generation-quality loss. The presented integrated optimization solution enables local offline AIGC text-to-image generation on resource-constrained edge-side devices and delivers a feasible technical pathway for deploying multi-modal generative-AI applications on edge-hardware platforms.

Keywords: AIGC; Text-to-image; Diffusion model; Lightweight model; Edge-side deployment; Model pruning; Model quantization; Knowledge distillation

1. Introduction

AIGC multi-modal generative-AI text-to-image diffusion models can generate high-quality visual outputs conditioned on natural-language text prompts, and are widely utilized for digital-art creation, industrial product design, and educational-content-production scenarios [1]. Open-source generative models represented by Stable Diffusion contain large-size network backbones. The U-Net denoising network occupies the majority of total parameters; conventional inference workflows demand 50-100 iterative reverse-denoising sampling steps and heavily rely on high-performance GPU computing resources. Therefore, most real-world deployments are located on remote cloud GPU servers.

Cloud-deployed AIGC text-to-image services exhibit several non-negligible drawbacks: network-transmission latency exists during online inference; uploading user prompts and intermediate generation data toward cloud servers introduces potential user-privacy leakage risks; generative functions become completely unavailable under offline no-network conditions. If local AIGC text-to-image generation can be realized on resource-limited edge-side hardware such as smartphones and embedded development boards, it can support offline generation capabilities, protect end-user data privacy, and reduce cloud-server operational costs, thus possessing considerable practical application value.

However, edge-side devices are subject to strict hardware constraints including limited memory capacity, restricted computing power, and tight power-budget limits. Directly deploying vanilla Stable Diffusion onto edge-side hardware easily causes memory overflow, extremely slow inference speed, and poor practical usability. Existing lightweight-diffusion-model research can be roughly classified into three technical directions: model pruning, model quantization, and few-step-sampling knowledge distillation. Nevertheless, adopting any single optimization approach independently will bring severe image-quality degradation. Naive unselective pruning may damage multi-scale feature-extraction capacity of U-Net skip-connection pathways. Global uniform quantization causes severe feature-information loss for sensitive network layers. Pure few-step distillation often results in blurry generated textures and distorted object structures [2].

Related prior-art analysis: Li et al. [3] investigated diffusion-model structured pruning by using magnitude-based non-selective pruning criteria, without special protection for U-Net skip-connection modules, which leads to obvious visual-quality degradation. Chen et al. [4] studied global INT8 quantization for diffusion-model networks; high-sensitivity layers suffer substantial feature loss under unified low-bit quantization. Zhang [5] explored few-step-sampling distillation for diffusion models, yet pruning-quantization compression is not incorporated, and model parameter volume remains too large for edge-side practical deployment.

Existing survey papers summarize general challenges for edge-side AIGC, but integrated end-to-end lightweight pipelines combining pruning, differentiated quantization and distillation are still insufficient.

Research gap: Integrated all-in-one lightweight-optimization pipelines for resource-constrained edge-side AIGC text-to-image diffusion models remain insufficient. Few existing works collaboratively accomplish structured pruning, differentiated mixed-precision quantization and few-step knowledge distillation while mitigating image-generation-quality regression artifacts.

Main contributions of this paper:

1. Importance-aware structured-channel pruning targeting diffusion-model U-Net networks: convolution-channel importance scores are calculated to evaluate each channel's contribution toward final generation outputs; sensitive skip-connection and output layers are explicitly protected from pruning operations to avoid erroneous removal of critical feature channels.

2. Differentiated mixed-precision-quantization strategy: different quantization bit-widths are assigned for individual network layers according to per-layer sensitivity characteristics. High-sensitivity layers retain FP16 floating-point precision; ordinary intermediate layers adopt INT8 quantization; low-importance layers apply INT4 quantization, balancing model storage-volume and image-generation visual quality.

3. Few-step-sampling knowledge distillation: a 50-step high-quality teacher diffusion-model supervises an 8-step lightweight student model, drastically decreasing reverse-denoising iteration overhead for edge-side inference.

4. Edge-aware multi-component joint-loss function is constructed, integrating MSE pixel loss, intermediate feature-matching loss and Inception-based perceptual loss, to suppress image-distortion and detail-loss artifacts introduced by pruning-quantization-distillation combined optimization.

5. Comparative experiments and ablation studies are carried out on the COCO-2014 public text-to-image dataset. Model-parameter size, inference runtime, FID and IS evaluation metrics are measured under simulated edge-CPU-hardware conditions to validate the overall effectiveness of the proposed integrated optimization scheme.

The rest of this paper is structured as follows. Section 2 reviews related background theories. Section 3 describes the complete LW-Diffusion lightweight-optimization pipeline. Section 4 introduces experimental datasets, hardware-software setup and evaluation metrics. Section 5 presents experimental results and ablation-study analysis. Section 6 discusses model limitations

and promising future-work directions. Section 7 draws the conclusion. References are appended at the end.

2. Related Work

2.1 Stable-Diffusion latent diffusion model

Stable Diffusion belongs to the latent-diffusion-model family and executes diffusion-denoising operations inside compressed latent space [1]. Its core functional components include the CLIP text encoder, U-Net denoising network, and VAE variational auto-encoder. The U-Net module consumes the dominant proportion of total computational overhead and contains massive convolution layers together with multi-scale skip-connection pathways. During inference-time reverse-denoising sampling, larger sampling-step numbers generally improve visual generation quality but increase inference runtime linearly. Two primary bottlenecks restricting edge-side deployment are the huge parameter volume of the U-Net backbone and the large quantity of reverse-sampling iterations.

2.2 Structured model pruning

Structured channel-level pruning removes redundant convolution channels as a unit, directly reducing model parameter count and computational complexity. Unstructured pruning produces sparse weight matrices, which demonstrate poor acceleration compatibility for common edge-side CPU/NPU hardware. Hence this paper selects structured channel-pruning for better hardware-execution adaptability. One major challenge for U-Net pruning is that skip-connection pathways transmit multi-scale critical feature information; blind unselective pruning will trigger severe visual-generation artifacts.

2.3 Model quantization

Quantization maps high-precision floating-point weight values into low-bit integer representations to reduce memory footprint and accelerate inference execution. Global uniform quantization yields unsatisfactory performance for diffusion-model networks, since certain U-Net layers are highly sensitive to quantization noise and will produce image-distortion artifacts after aggressive low-bit quantization. Mixed-precision quantization assigns different bit-width configurations for different layers to achieve a trade-off between inference speed and generated-image visual quality.

2.4 Diffusion-model knowledge distillation and few-step sampling

Knowledge distillation transfers generative knowledge from a high-quality high-step teacher

model toward a lightweight student model; the student model accomplishes image generation with far fewer reverse-denoising sampling steps. Conventional few-step distillation schemes merely adopt pixel-level MSE loss and easily suffer texture-detail loss; additional perceptual-loss constraints are required.

2.5 Core practical challenges for edge-side AIGC text-to-image deployment

1. **Memory constraints**: Vanilla Stable Diffusion consumes several gigabytes of memory resources, while embedded-device RAM capacity is strictly limited.
2. **Computational constraints**: Most edge-side hardware lacks high-performance GPU units and only relies on CPU or lightweight NPU computing resources.
3. **Power-budget constraints**: Battery-powered mobile devices cannot sustain heavy computational power consumption during inference.
4. **Visual-quality constraints**: After lightweight optimization operations, artifacts including object structural distortion and texture blurring should be suppressed as much as possible.

Primary evaluation indicators:

- FID (Fréchet Inception Distance): Smaller FID values indicate higher similarity between generated-image distribution and real-image ground-truth distribution.
- IS (Inception Score): Larger IS values represent better overall generated-image quality.
- Model parameter size, single-image inference runtime.

3. Methodology: LW-Diffusion Lightweight-Optimization Pipeline

The integrated LW-Diffusion optimization workflow is as follows: Original Stable Diffusion v1.5 model → **importance-aware structured-channel pruning (U-Net sensitive skip-connection layer protection) → differentiated mixed-precision quantization → few-step-sampling knowledge-distillation training (teacher model 50-step, student model 8-step) → edge-aware multi-component joint-loss-function constrained training** → output final LW-Diffusion lightweight edge-deployable model.

3.1 Importance-aware structured-channel pruning

U-Net skip-connection convolution layers and final output convolution layers are classified as high-sensitivity layers; pruning channels belonging to these layers will induce severe image-distortion artifacts. The detailed pruning workflow is described below:

1. Traverse the convolutional channels in the U-Net and compute a channel-importance score

that estimates each channel's contribution to the denoising output.

AUTHOR ACTION REQUIRED: Provide the exact mathematical definition of the channel-importance score and explain how it is computed. This is essential for reproducibility and for substantiating the "importance-aware" contribution.

2. Construct the sensitive-layer-protection list: U-Net skip-connection convolutions and final-output convolution layers are added to this list; channels inside protected layers are strictly prohibited from being pruned.

3. For ordinary non-protected convolution layers, sort convolution channels according to pre-computed importance scores and prune low-importance redundant channels; a global pruning-ratio upper bound is enforced to avoid excessive aggressive pruning.

4. Fine-tune the pruned network to repair parameter perturbations introduced by channel-removal operations.

> Pruning pseudocode

```
Plain Text
Input: Original U-Net network, pruning-rate P_rate, Protect_Layer (sensitive-layer-protection list)
Output: Pruned network U-Net_pruned
1: Compute Importance_score for every convolution channel
2: for each conv layer in U-Net:
3:      if conv ∈ Protect_Layer: skip, no pruning
4:      else: sort channels by Importance_score, prune low-score redundant channels
5: Execute fine-tuning to repair pruning-induced parameter perturbation
6: return U-Net_pruned
```

3.2 Differentiated mixed-precision quantization

After structured pruning, mixed-precision quantization is applied. Precision is assigned according to the layer-sensitivity criterion used in the implementation, with critical layers retained at higher precision and less-sensitive layers quantized more aggressively.

AUTHOR ACTION REQUIRED: Define the layer-sensitivity metric/procedure and report which layers (or proportions of layers/parameters) use FP16, INT8, and INT4. State whether quantization is post-training or quantization-aware and identify the

implementation/backend.

1. **High-sensitivity layers (U-Net output-layer, skip-connection-related convolution layers)**: Retain FP16 high-precision representation and avoid feature corruption caused by low-bit quantization.
2. **Medium-sensitivity ordinary convolution layers**: Adopt INT8 quantization.
3. **Low-sensitivity deep intermediate convolution layers**: Apply INT4 quantization to maximize storage-volume compression ratio.

Instead of adopting global uniform INT8 or INT4 quantization, differentiated mixed-precision quantization sacrifices precision for less-important intermediate layers while maximally preserving feature information carried by critical network layers, alleviating image-detail-loss artifacts originating from quantization operations.

3.3 Few-step-sampling knowledge distillation

- **Teacher model**: Original Stable Diffusion v1.5 model, inference sampling steps 50, generates high-quality reference images.
- **Student model**: Pruned plus mixed-quantized LW-Diffusion target lightweight model, target inference sampling steps 8.

Distillation workflow: Input text prompts are fed into both models. The teacher model executes complete 50-step reverse-diffusion sampling and outputs high-quality teacher-reference images. The student lightweight model generates outputs with merely 8-step reverse-denoising sampling. Teacher-model generation outputs are utilized as supervision signals for training the student model, transferring generative knowledge from the high-step teacher model toward the few-step student model.

3.4 Edge-aware multi-component joint-loss function

Pure pixel-level MSE loss can encourage overly smooth outputs. The proposed training objective therefore combines pixel-level reconstruction, intermediate feature matching, and a perceptual component:

1. L_{mse}: Pixel-wise mean-square-error loss between student-model generated images and teacher-model reference images.
2. L_{feat}: Intermediate-feature-matching loss, aligning intermediate-layer feature outputs of teacher model and student model to preserve texture details.

3. $L_{fid_perceptual}$: Inception-network-based perceptual loss, constraining high-level semantic distribution and suppressing object-distortion artifacts.

The weighting coefficients α and β are tuned on a held-out validation set.
AUTHOR ACTION REQUIRED: Insert the actual α and β values, tuning range/procedure, and validation criterion used to select them.

3.5 Complete training-inference workflow

Training phase:

1. Perform importance-aware structured-channel pruning on the original model, with U-Net sensitive-layer protection enabled.
2. Apply differentiated mixed-precision quantization for network weight parameters.
3. Load the pre-trained 50-step teacher diffusion-model, execute few-step knowledge-distillation training; back-propagate and update student-model parameters guided by the multi-component joint-loss function.
4. Iterate until training converges to obtain the final LW-Diffusion lightweight edge-side text-to-image AIGC model.

Edge-side inference phase:

1. Input text prompt; the CLIP text-encoder computes text embedding vectors.
2. LW-Diffusion U-Net plus VAE modules execute 8-step reverse-denoising sampling.
3. The VAE decoder outputs final generated images.

In the reported experiments, inference is executed locally in a CPU-only environment without cloud-server access. These results provide a simulation-based indication of the feasibility of offline deployment on resource-constrained platforms; validation on actual mobile or embedded edge hardware remains necessary.

> **Figure 2 placeholder**: Overall pipeline diagram of LW-Diffusion optimization workflow. Insert actual figure here.

4. Experiments

4.1 Dataset

Experiments use 50,000 text-image pairs sampled from the MS COCO 2014 corpus. Text

descriptions are used as input prompts and the corresponding images as reference images. The train/validation/test construction should follow the protocol used in the actual experiment.
AUTHOR ACTION REQUIRED: Specify exactly which COCO 2014 split(s) and captions were used, how the 50,000 pairs were selected, and the exact train/validation/test sample counts. Do not describe COCO 2014 itself as a 50,000-pair dataset unless that is the precise experimental subset.

Evaluation metrics:
AUTHOR ACTION REQUIRED: Complete the experimental protocol before submission: final pruning rate, image resolution, sampler/scheduler, optimizer, learning rate, fine-tuning/training sample counts, random seed(s), number of runs, FID/IS computation protocol, and whether reported values are single runs or averages (with variability where applicable).

1. Model parameter size (M).
2. Single-image inference runtime (CPU-only environment, unit: second).
3. FID (Fréchet Inception Distance; lower value corresponds to better visual quality).
4. IS (Inception Score; higher value corresponds to better visual quality).

Comparative baseline schemes:

- Baseline 1: Vanilla Stable Diffusion v1.5 (50-step sampling).
- Baseline 2: Only global undifferentiated structured pruning.
- Baseline 3: Only global INT8 quantization.
- Baseline 4: Pruning + global INT8 quantization (without sensitive-layer protection, without mixed-precision quantization).
- Proposed LW-Diffusion: importance-aware pruning (sensitive-layer protection) + mixed-precision quantization + few-step distillation + joint-perceptual loss.

Ablation experimental groups:
① Remove sensitive-layer protection; adopt naive undifferentiated pruning.
② Replace differentiated mixed-precision quantization with global uniform INT8 quantization.
③ Remove few-step distillation; retain original 50-step inference-sampling.
④ Full LW-Diffusion framework with all functional modules enabled.

5. Experimental Results and Analysis

5.1 Overall comparative experimental results under simulated edge-CPU environment

Table 4 lists overall quantitative comparison outcomes for all competing approaches.

Table 4. Overall quantitative comparison results under simulated edge-CPU environment

Model	Parameters (M)	Inference Time (s)	FID	IS
Vanilla Stable Diffusion v1.5	983.2	28.7	19.42	23.76
Only global naive pruning	446.5	14.3	34.85	16.21
Only global INT8 quantization	724.1	16.5	29.14	18.43
Pruning + global INT8 quantization	352.4	8.6	31.26	17.05
LW-Diffusion (Proposed)	**311.7**	**4.2**	**23.67**	**21.04**

LW-Diffusion compresses total model parameters down to 31.7 % of the original Stable Diffusion v1.5 model. Under pure-CPU simulation conditions, average single-image inference runtime reaches 4.2 s, which satisfies basic usability requirements for edge-side lightweight applications. FID equals 23.67, representing moderate visual-quality regression compared against the vanilla full-model baseline, yet substantially outperforms other standalone- or simple-combined lightweight baselines. Naive unselective pruning and global uniform quantization induce severe FID deterioration and frequently produce object-distortion and texture-corruption visual artifacts. The proposed sensitive-layer-protection mechanism, differentiated mixed-precision quantization and multi-component joint-loss collaboratively suppress visual-quality degradation originating from lightweight-optimization operations.

5.2 Ablation-study experimental results

Ablation experiments are conducted to validate the individual contribution of each core functional component, as shown in Table 5.

Table 5. Ablation-study experimental results

Experimental Setup	Parameters (M)	FID	Inference Time (s)	Explanation
No sensitive-layer protection, naive pruning	304.3	30.41	4.0	Critical skip-connection layers get pruned → severe image distortion
Pruning + global INT8 quantization, no mixed-precision	308.5	27.83	4.1	High-sensitivity layers suffer INT8 quantization → heavy detail loss
Pruning + mixed-precision, remove few-step distillation (50-step inference)	314.2	21.54	17.4	Acceptable image quality, but inference is too slow for edge-side deployment
Full LW-Diffusion framework	311.7	23.67	4.2	Optimal comprehensive trade-off among parameter-size, inference-speed and visual generation quality

Ablation-study outcomes demonstrate that the U-Net sensitive-skip-connection-protection mechanism, differentiated mixed-precision quantization, and few-step knowledge-distillation module are all indispensable functional components. Omitting any single module will cause obvious visual-quality degradation or unsatisfactory edge-side inference-speed performance.

5.3 Qualitative generated-sample analysis

Visual comparison among different groups of generated samples: The vanilla full-model baseline produces outputs with clear object contours and rich fine-grained textures. Baselines adopting naive pruning or global uniform quantization frequently generate outputs containing distorted object structures, broken limbs and smeared textures. Although minor fine-detail loss exists for LW-Diffusion compared to the vanilla full-model baseline, object overall structures remain complete without obvious severe distortion artifacts, which makes it suitable for edge-side lightweight AIGC-generation application scenarios.

> **Figure 3 placeholder**: Qualitative visual-sample comparison sub-figure group.
> Insert actual generated-image comparison figure here.

5.4 Experiments for different pruning-ratio configurations

As the pruning ratio increases, parameter count decreases and inference speed improves, while FID deteriorates. In the reported experiments, a pruning ratio of 0.62 is identified as the selected trade-off point; degradation becomes more pronounced when the pruning ratio exceeds 0.70. **AUTHOR ACTION REQUIRED: Add the actual pruning-ratio experiment table/curve and state the tested ratios, selection criterion, and corresponding parameter count, runtime, FID and IS values. Without these data, the claim that 0.62 is optimal is not independently supported by the manuscript.**

6. Discussion

Limitations of the proposed LW-Diffusion scheme:

1. Compared with the original vanilla Stable Diffusion v1.5 full-model, LW-Diffusion still brings moderate visual-quality regression; it represents a necessary speed-quality trade-off. This scheme targets lightweight offline edge-side AIGC application scenarios. Professional high-fidelity artistic-creation scenarios still prefer cloud-deployed full-capacity diffusion-model variants.

2. This paper mainly focuses on text-to-image generation tasks. Further adaptation and parameter fine-tuning are required for derivative diffusion-model tasks including image

inpainting and image out-painting.

3. Our main experiments are executed under CPU-simulated environment. Additional operator-level adaptation work is required for practical deployment on real-world mobile-NPU and embedded-hardware platforms.

Future research directions:

1. Combine LoRA low-rank-adaptation technology to implement lightweight few-shot style fine-tuning for edge-side AIGC models.

2. Further optimize distillation-loss design to narrow the visual-quality gap between lightweight student model and original full-capacity teacher model.

3. Carry out operator-adaptation engineering and real-device validation targeting genuine mobile-terminal NPU hardware, to realize practical end-user-oriented edge-side AIGC text-to-image applications.

7. Conclusion

This paper addresses the computational and memory demands that hinder deployment of text-to-image diffusion models on resource-constrained platforms. LW-Diffusion integrates importance-aware structured channel pruning with protected U-Net skip-connection layers, differentiated mixed-precision quantization, few-step knowledge distillation, and a multi-component training loss. On the reported COCO-2014 experimental subset, the model retains 31.7% of the parameter count of the Stable Diffusion v1.5 baseline, reduces sampling from 50 to 8 steps, and achieves a 4.2 s average inference time in the reported CPU-only simulation, with an FID of 23.67. These findings indicate a promising speed-size-quality trade-off under the evaluated setting, while real-device mobile/NPU and embedded-hardware validation remains necessary before practical edge deployment can be established.

References

AUTHOR ACTION REQUIRED: Verify every reference against the original publication and expand the literature review with recent, directly relevant work on diffusion-model acceleration, few-step/consistency distillation, diffusion quantization/pruning, and mobile/edge diffusion deployment. Ensure all bibliography entries follow the target journal style.

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